

68. Apple Authentication 3.0 Coprocessor

68.1 Overview

Apple Authentication 3.0 Coprocessor (Auth 3.0 CP) is used to authenticate an accessory with devices running iOS 10.0.2 or later. See [Accessory Authentication](#) (page 211) for more information.

The Auth 3.0 CP is available in the following versions:

- [XDFN Package](#) (page 593).
- [WLCSP Package](#) (page 595).

See [Table 93-1](#) (page 754) for sample part numbers.

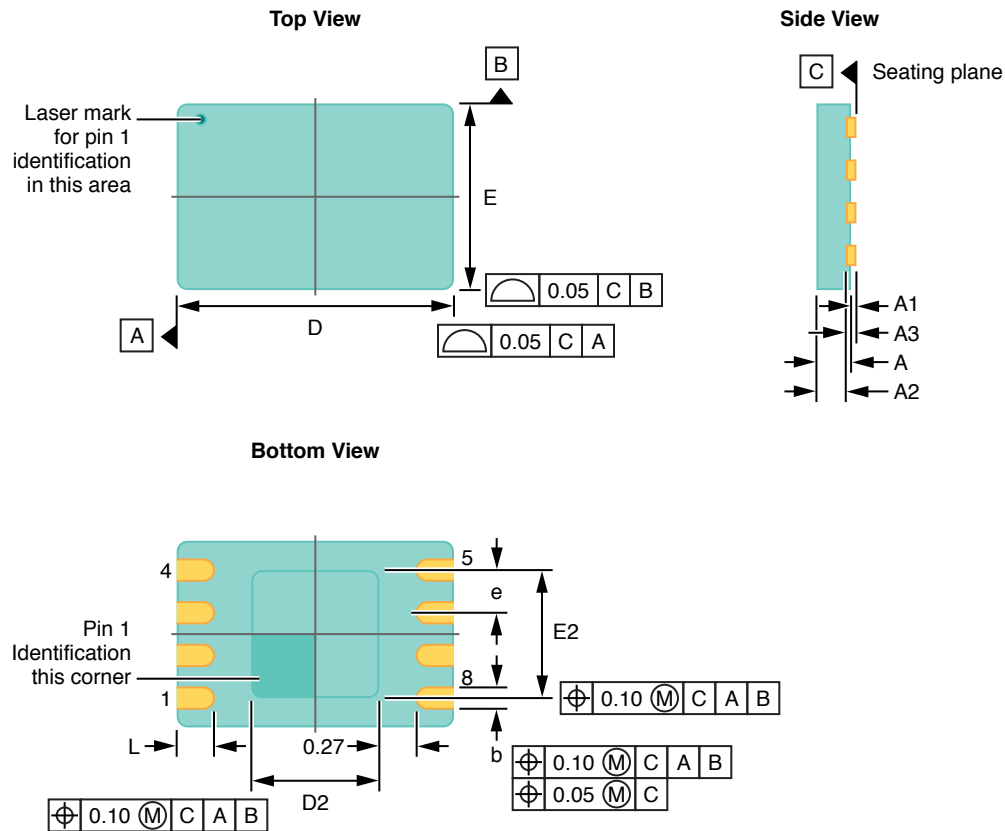
The Auth 3.0 CP Export Classification Control Number (ECCN) is EAR99. Both XDFN and WLCSP versions have a moisture sensitivity level (MSL) of 1.

68.2 Mechanical

This section provides technical details and tolerances for the Auth 3.0 CP chip.

68.2.1 XDFN Package

[Figure 68-1](#) (page 594) shows the Auth 3.0 CP's package's layout, pad locations, and dimensional tolerances.

Figure 68-1 Auth 3.0 CP package (XDFN)**Table 68-1** Auth 3.0 CP dimensions (XDFN)

Symbol	Minimum	Nominal	Maximum
A			0.40 mm
A1			0.05 mm
A2		0.23 mm	0.28 mm
A3		0.127 mm (REF)	
b	0.18 mm	0.25 mm	0.30 mm
D		3.00 mm (BSC)	
D2	1.56 mm	1.66 mm	1.76 mm
E		2.00 mm (BSC)	
E2	1.40 mm	1.50 mm	1.60 mm
L	0.35 mm	0.40 mm	0.45 mm
e		0.50 mm (BSC)	

Figure 68-2 (page 595) and Table 68-2 (page 595) show the Auth 3.0 CP's recommended land pattern dimensions. The package includes a central paddle (exposed pad) that may be left not connected or may be connected to Ground.

Figure 68-2 Auth 3.0 CP land pattern dimensions (XDFN)

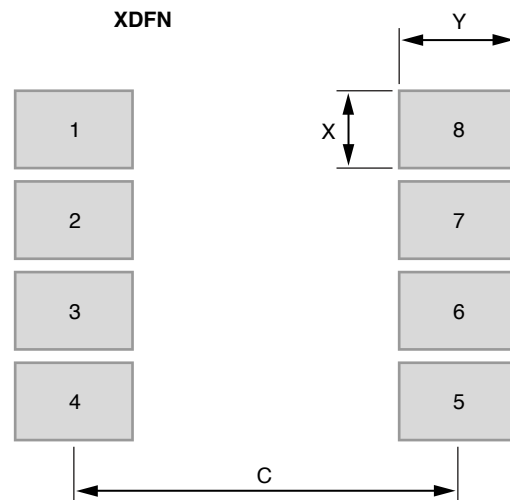
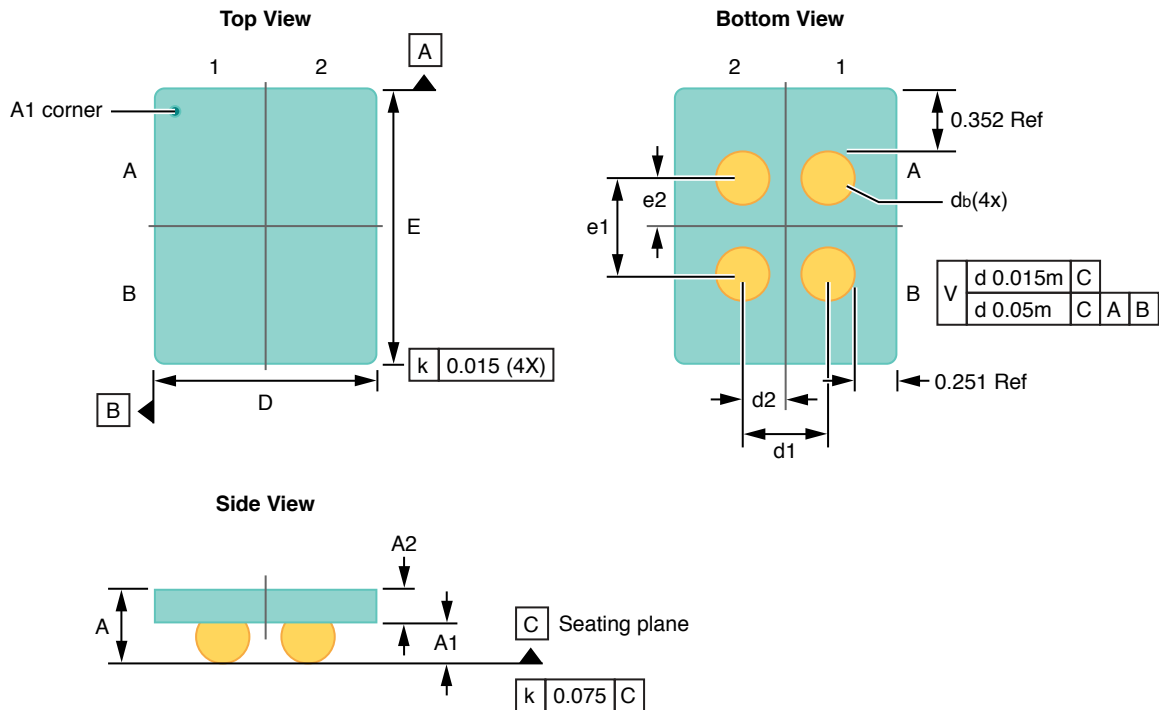


Table 68-2 Auth 3.0 CP land pattern dimensions (XDFN)

Dimension/Symbol	Value
Body	2 mm x 3 mm
Pitch	0.50 mm
Lead Length	0.45 mm
C	2.90 mm
X	0.30 mm
Y	0.90 mm

68.2.2 WLCSP Package

Figure 68-3 (page 596) shows the Auth 3.0 CP's package's layout, pad locations, and dimensional tolerances.

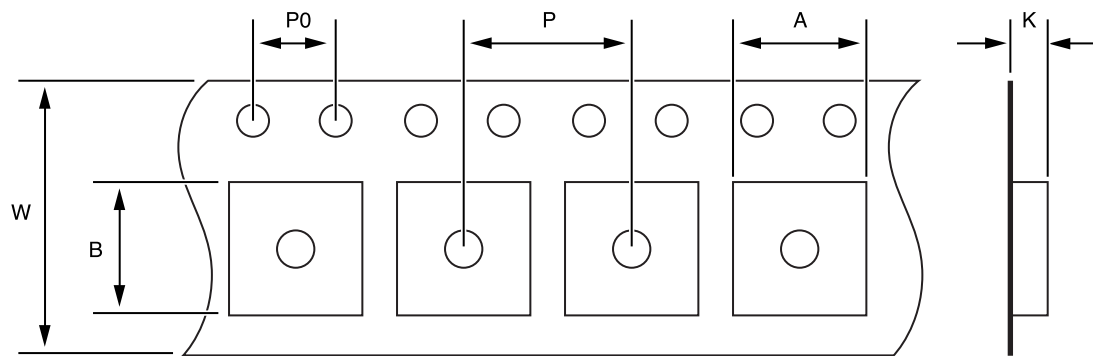
Figure 68-3 Auth 3.0 CP package (WLCSP)**Table 68-3** Auth 3.0 CP dimensions (WLCSP)

Symbol	Minimum	Nominal	Maximum
A	0.330 mm	0.360 mm	0.390 mm
A1	0.160 mm	0.175 mm	0.190 mm
A2	0.170 mm	0.185 mm	0.200 mm
D	1.122 mm	1.152 mm	1.182 mm
d1	0.400 mm (BSC)		
d2	0.200 mm (BSC)		
E	1.424 mm	1.454 mm	1.484 mm
e1	0.500 mm (BSC)		
e2	0.250 mm (BSC)		
b	0.230 mm	0.250 mm	0.270 mm

The ball shear is 161 g/ball. Apple recommends mounting Auth 3.0 CP on a PCB with 227 μ m diameter round pads.

68.2.3 Packing

Figure 68-4 (page 597) and Figure 68-5 (page 598) describe how the Auth 3.0 CP is delivered to accessory manufacturing facilities.

Figure 68-4 Auth 3.0 CP packing tape**Table 68-4** Auth 3.0 CP packing tape dimensions (XDFN)

Symbol	Minimum	Nominal	Maximum
P0	3.90 mm	4.00 mm	4.10 mm
P	3.90 mm	4.00 mm	4.10 mm
A	2.15 mm	2.20 mm	2.25 mm
B	3.15 mm	3.20 mm	3.25 mm
K	0.48 mm	0.53 mm	0.58 mm
W	11.90 mm	12.00 mm	12.30 mm

Table 68-5 Auth 3.0 CP packing tape dimensions (WLCSP)

Symbol	Minimum	Nominal	Maximum
P0	3.90 mm	4.00 mm	4.10 mm
P	1.95 mm	2.00 mm	2.05 mm
A	1.24 mm	1.29 mm	1.34 mm
B	1.54 mm	1.59 mm	1.64 mm
K	0.44 mm	0.49 mm	0.54 mm
W	7.90 mm	8.00 mm	8.30 mm

Figure 68-5 Auth 3.0 CP packing reel (XDFN and WLCSP)

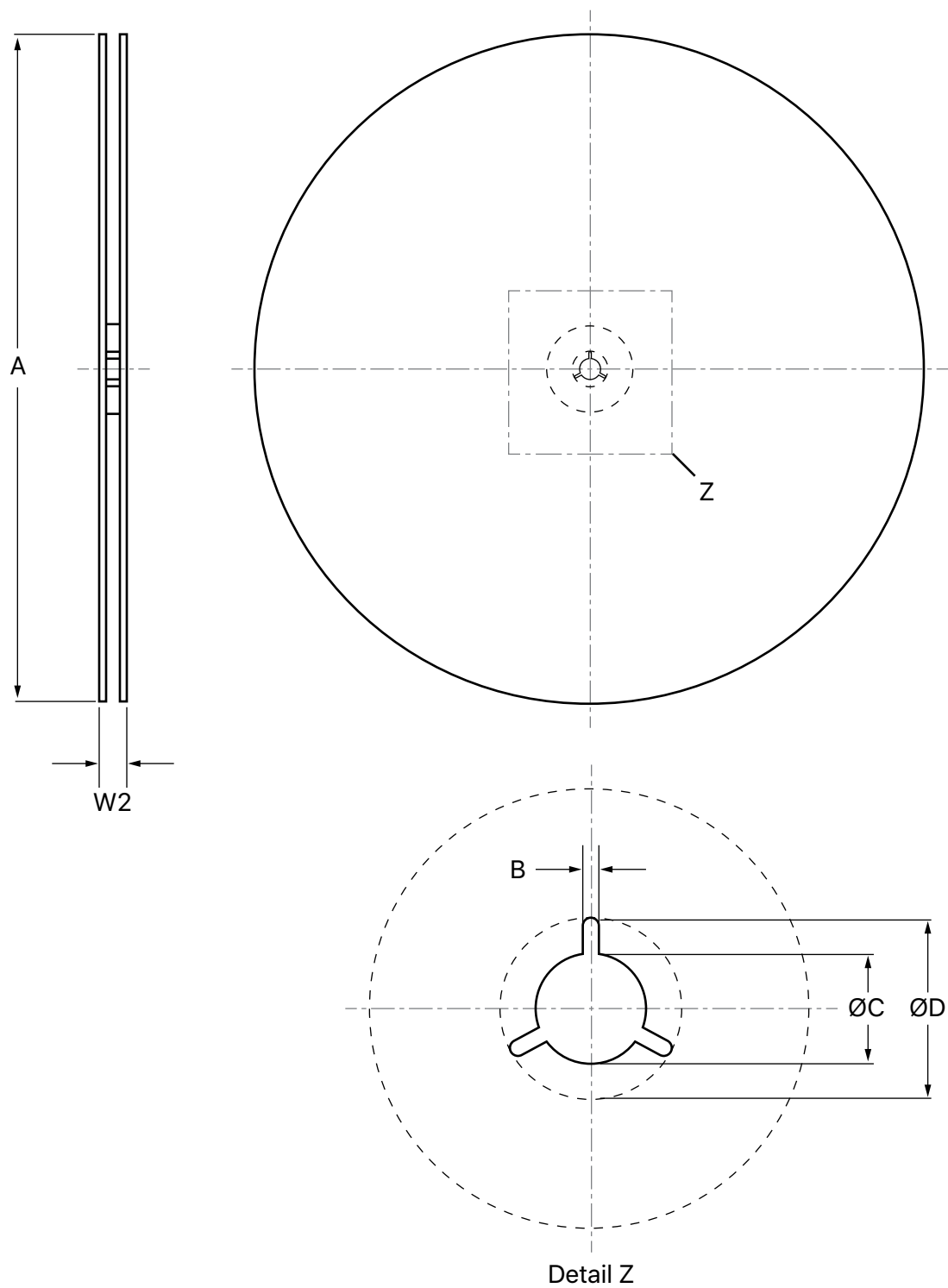


Table 68-6 Auth 3.0 CP packing reel dimensions (XDFN)

Symbol	Minimum	Nominal	Maximum
A	328.0 mm	330.0 mm	332.0 mm
W2			18.4 mm
B	1.5 mm	2.0 mm	2.5 mm
ØC	12.8 mm	13.0 mm	13.5 mm
ØD	20.2 mm		

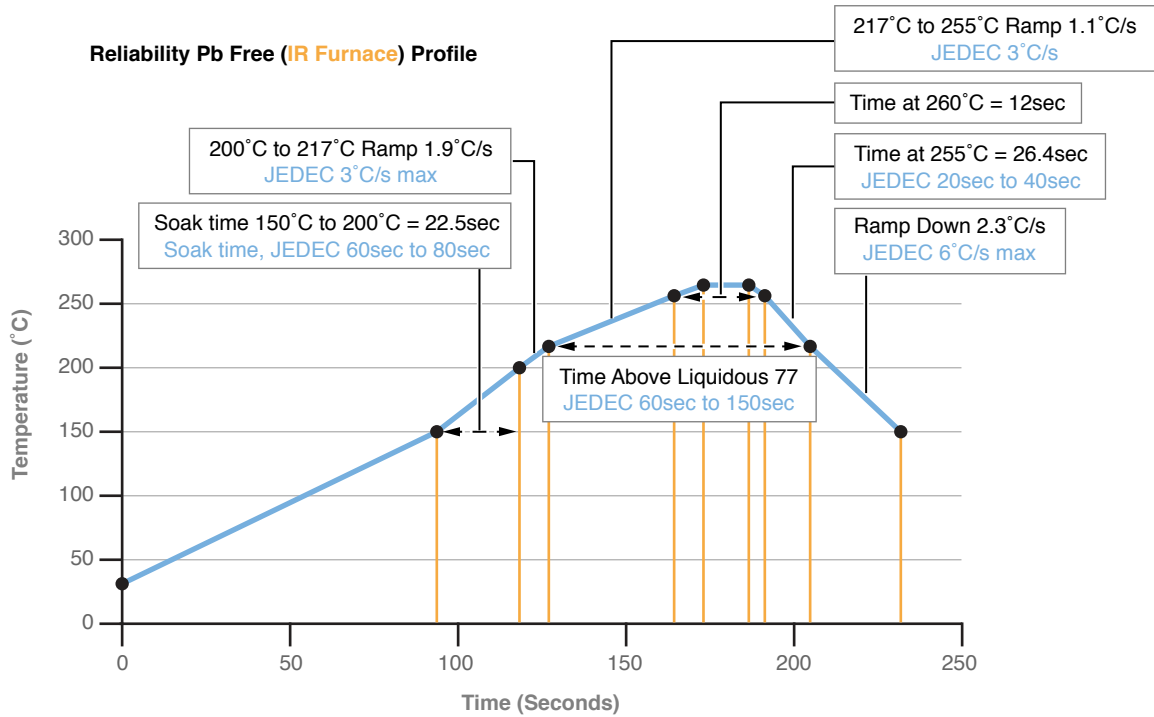
Table 68-7 Auth 3.0 CP packing reel dimensions (WLCSP)

Symbol	Minimum	Nominal	Maximum
A	328.0 mm	330.0 mm	332.0 mm
W2			14.4 mm
B	1.5 mm	2.0 mm	2.5 mm
ØC	12.8 mm	13.0 mm	13.5 mm
ØD	20.2 mm		

68.2.4 Handling

The Auth 3.0 CP package is Pb-free and is designed to withstand a standard reflow profile with a peak reflow temperature below 260 °C. See [Figure 68-6](#) (page 600) for the reflow profile of the Auth 3.0 CP (XDFN and WLCSP).

Figure 68-6 Auth 3.0 CP reflow profile (XDFN and WLCSP)



68.3 Pad Assignments

68.3.1 XDFN Package

Figure 68-7 Auth 3.0 CP, top view (XDFN)

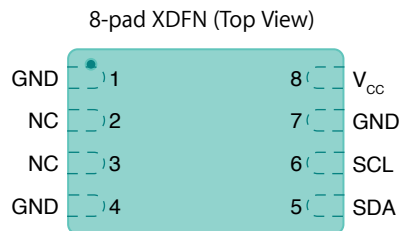


Table 68-8 Auth 3.0 CP signals (XDFN)

Signal	Pad	I/O	Description
GND	1		Ground
NC	2-3		Must not be connected.

Signal	Pad	I/O	Description
GND	4		Ground
SDA	5	I/O	I ² C data
SCL	6	I	I ² C clock
GND	7	I	Ground
V _{CC}	8		Supply voltage

Note: Pads 1, 4, and 7 are internally connected to each other. The ground terminal of the bypass capacitor should be as close as possible to any one of these three pads.

68.3.2 WLCSP Package

Figure 68-8 Auth 3.0 CP, top view (WLCSP)

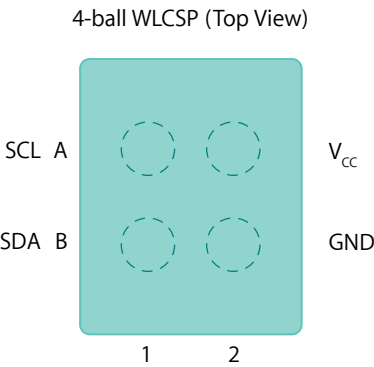


Table68-9 Auth 3.0 CP signals (WLCSP)

Signal	Pad	I/O	Description
SCL	1A	I	I ² C clock
SDA	1B	I/O	I ² C data
V _{CC}	2A		Supply voltage
GND	2B		Ground

68.4 Electrical

This section provides technical details and tolerances for the Auth 3.0 CP chip.

68.4.1 Maximum Environmental Conditions

Table 68-10 (page 602) lists the Auth 3.0 CP's absolute maximum electrical and free-air temperature ranges. Stresses to the Auth 3.0 CP beyond the ranges may cause permanent damage. Exposure to either end of any range may affect the Auth 3.0 CP reliability.

Table 68-10 Auth 3.0 CP maximum electrical and temperature ranges

Condition	Maximum Range
Voltage applied at V_{CC} relative to V_{SS}	-0.3 V to +6.0 V
Voltage applied to any pad	-0.3 V to $V_{CC} + 0.3$ V
Storage temperature	-65 to +150 °C

68.4.2 Recommended Operating Conditions

Internal sensors may force the Auth 3.0 CP to its reset state if any of the conditions listed in **Table 68-11** (page 602) are exceeded. Attempting to operate the Auth 3.0 CP in this state is not recommended and may lead to the Auth 3.0 CP failure or unreliability.

Table 68-11 Auth 3.0 CP maximum electrical and temperature ranges

Condition	Recommended Range
Operating free-air temperature	-40 to +85 °C
Supply voltage	1.71 V to 3.6 V

68.4.3 ESD Characteristics

Table 68-12 ESD characteristics

Pad	Description	ESD Requirement	Voltage
V_{CC}	Supply voltage	CDM ± 1 kV, HBM ± 4 kV	+3.6 V
SDA	I ² C data	IEC 61000-4-2 ± 8 kV Contact Discharge	+8.0 V limited to 150 ms max over lifetime.
SCL	I ² C clock	CDM ± 1 kV, HBM ± 4 kV	+5.5 V

68.4.4 DC Characteristics

Table 68-13 DC characteristics

Item	Description	Minimum	Typical	Maximum
I_{CHAL}	Operating current while computing challenge	-	-	1.15 mA
I_{SLEEP}	Sleep current	-	100 nA	2000 nA
V_{OL}	Output Voltage Low, $I_{\text{OL}} = 8 \text{ mA}$	-	-	0.4 V
V_{OL}	Output Voltage Low, $I_{\text{OL}} < 500 \text{ }\mu\text{A}$	-	-	0.2 V
C_{SDA}	Input capacitance of SDA	-	60 pF	-
C_{SCL}	Input capacitance of SCL	-	3 pF	-
V_{CC}	Supply voltage	1.71 V	-	3.6 V
I_{IH}	Input Current ($V_{\text{IH}} = 0 \text{ V}, 3.6 \text{ V}$)	-1 μA	-	1 μA
V_{IH}	Input Voltage High	$0.7 \text{ V} \times V_{\text{CC}}$	-	3.9 V
V_{IL}	Input Voltage Low	-0.3 V	-	$0.3 \text{ V} \times V_{\text{CC}}$

68.4.5 AC Characteristics

This section is applicable over recommended operating range from $T_{\text{A}} = -40^{\circ}$ to $+85^{\circ}$, $V_{\text{CC}} = +1.71 \text{ V}$ to $+3.6 \text{ V}$.

Table 68-14 AC characteristics

Item	Description	Minimum	Typical	Maximum
t_{WATCHDOG}	Watchdog Timeout to drop into sleep	3.2 s	4 s	4.8 s
t_{DIT}	Auth 3.0 CP Initialization Time from power applied or wake from sleep	-	-	1 ms
t_{AUTH}	Authentication Time	-	-	500 ms
t_{CERT}	Certificate Generation	-	-	10 ms
t_{READ}	EEPROM Reads	-	-	1.5 ms
t_{WRITE}	EEPROM Writes	-	-	20 ms
t_{SN}	Serial Number Formatting	-	-	1 ms
t_{CMD}	All Other Commands	-	-	800 μs
f_{SCK}	SCL Clock Frequency	0 kHz	-	400 kHz

Item	Description	Minimum	Typical	Maximum
t_{HIGH}	SCK High Time	0.6 μs	-	-
t_{LOW}	SCK Low Time	1.2 μs	-	-
$t_{\text{SU.STA}}$	Start Setup Time	0.6 μs	-	-
$t_{\text{HD.STA}}$	Start Hold Time	0.6 μs	-	-
$t_{\text{SU.STO}}$	Stop Setup Time	0.6 μs	-	-
$t_{\text{SU.DAT}}$	Data In Setup Time	100 ns	-	-
$t_{\text{HD.DAT}}$	Data In Hold Time	0 ns	-	-
t_{R}	Input Rise Time from $0.1 \times V_{\text{CC}}$ to $0.9 \times V_{\text{CC}}$	-	-	1 μs
t_{F}	Input Fall Time from $0.9 \times V_{\text{CC}}$ to $0.1 \times V_{\text{CC}}$	-	-	1 μs
t_{AA}	Clock Low to Data Out Valid	-	-	0.9 μs
t_{DH}	Data Out Hold Time	50 ns	-	-
t_{BUF}	Time bus must be free before a new transmission can start.	1.2 μs	-	-
t_{I}	Input Filter Glitch Suppression (SCL,SDA)	-	-	50 ns

Note: Calculation time measured from the end of the trailing break after command input to the point in time when the Auth 3.0 CP begins transmitting the first bit of the response.

68.4.6 Sleep Mode

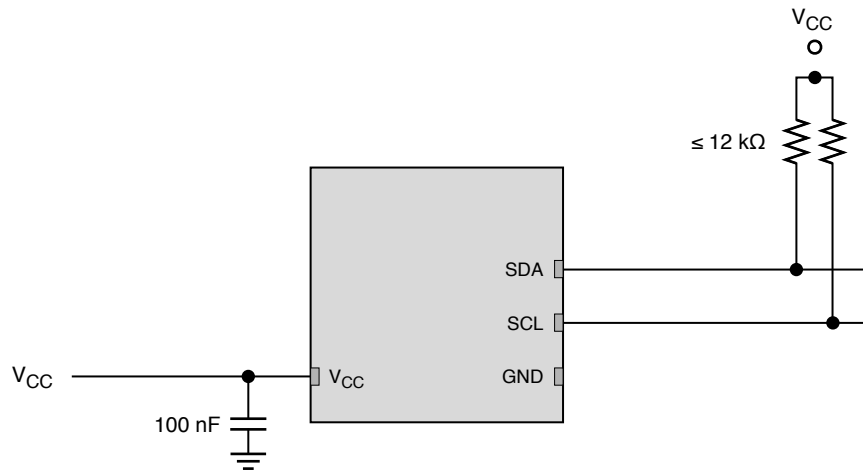
The Auth 3.0 CP conserves power by entering a sleep mode. It enters this mode automatically after the watchdog timer expires (t_{WATCHDOG}) or via the sleep register, see [Sleep](#) (page 614).

When in sleep mode, the Auth 3.0 CP automatically wakes in response to any I²C communications sent to its address.

68.4.7 Reference Design

Accessories must implement the reference design as shown in [Figure 68-9](#) (page 605).

Figure 68-9 Auth 3.0 CP reference design



68.5 I²C Interface

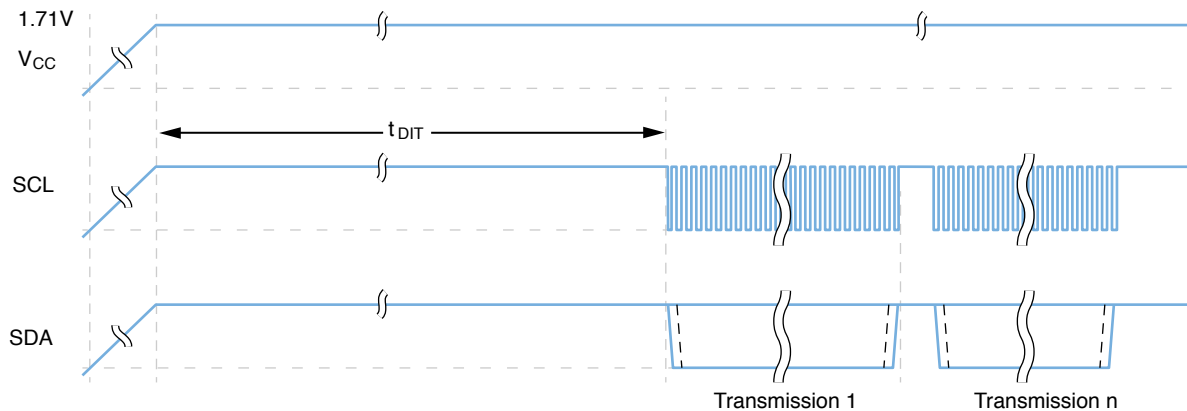
The Auth 3.0 CP acts as a I²C slave and uses a 7-bit slave address, see [Addressing](#) (page 606).

68.5.1 Setup

The I²C interface is started either by supplying power to the V_{CC} line or by any transition on the SDA line if the Auth 3.0 CP is in [Sleep Mode](#) (page 604).

The I²C output buffer is cleared on power-up and wake from sleep mode.

The Auth 3.0 CP requires a certain amount of time to read the I²C address when powered up. The first data transmission must start no earlier than t_{DIT} , see [Table 68-14](#) (page 603), after the V_{CC} line goes high. See [Figure 68-10](#) (page 606) for an I²C timing diagram.

Figure 68-10 Auth 3.0 CP I²C power up timing

68.5.2 Hardware

The I²C Interface operates at 100 kHz (standard mode) or 400 kHz (full speed) and consists of two signals:

- *SDA*: Data
- *SCL*: Clock

Accessories must connect pull-up resistors (12 k Ω maximum) from the same power supply as the Auth 3.0 CP to the SCL and SDA lines of the Auth 3.0 CP.

68.5.3 Addressing

The slave address for writing is shown in [Figure 68-11](#) (page 606) and the corresponding read address in [Figure 68-12](#) (page 606).

Figure 68-11 Auth 3.0 CP I²C slave write address

A6	A5	A4	A3	A2	A1	A0	R/nW
0	0	1	0	0	0	0	0

Figure 68-12 Auth 3.0 CP I²C slave read address

A6	A5	A4	A3	A2	A1	A0	R/nW
0	0	1	0	0	0	0	1

The I²C write address of the Auth 3.0 CP consists of the seven bits [A6:A0] followed by 0 for the R/nW bit. The I²C read address of the Auth 3.0 CP consists of the seven bits [A6:A0] followed by 1 for the R/nW bit. The write and read addresses of the Auth 3.0 CP are 0x20 and 0x21 respectively.

68.5.4 I²C Transaction Format

Table 68-15 Auth 3.0 CP read/write transaction data format

Field	Bytes	Description
I ² C Slave Address + R/W bit	1	Must be 0x20 for a write transaction or 0x21 for a read transaction.
Register	1	See Registers (page 608).
Payload	0–n	Payload data.

68.5.5 Writing to the Auth 3.0 CP

To write data to the Auth 3.0 CP, follow these steps:

1. Send the I²C start sequence.
2. Send the I²C write address of the Auth 3.0 CP.
3. Check for an ACK from the slave; if a NACK is received, wait 500 μs and then loop back to Step 1.
4. Send the register address at which to begin writing.
5. Check for an ACK from the slave; if a NACK is received, wait 500 μs and then loop back to Step 1.
6. Send the data bytes.
7. Send the I²C stop sequence.

The accessory must send a I²C stop sequence after the last data byte of any write to ensure that the Auth 3.0 CP will start the execution of the operation. The behavior of the Auth 3.0 CP is undefined if a I²C start sequence is sent prior to a I²C stop sequence.

68.5.6 Reading from the Auth 3.0 CP

To read data from the Auth 3.0 CP, follow these steps:

1. Send the I²C start sequence.
2. Send the I²C write address of the Auth 3.0 CP.
3. Check for an ACK from the slave; if a NACK is received, wait 500 μs and then loop back to Step 1.
4. Send the register address at which to begin reading.
5. Check for an ACK from the slave; if a NACK is received, wait 500 μs and then loop back to Step 1.
6. Send the I²C stop sequence.

7. Send the I²C start sequence.
8. Send the I²C read address of the Auth 3.0 CP.
9. Check for an ACK from the slave; if a NACK is received, wait 500 μ s and then loop back to Step 7.
10. Read the data bytes.
11. Send the I²C stop sequence.

If the read buffer on the accessory is limited and cannot read all the desired bytes prior to issuing the I²C stop sequence in Step 11, then the following sequence may be repeated as many times as necessary to retrieve all remaining bytes:

1. Send the I²C start sequence.
2. Send the I²C read address of the Auth 3.0 CP.
3. Check for an ACK from the slave; if a NACK is received, wait 500 μ s and then loop back to Step 6.
4. Read the data bytes.
5. Send the I²C stop sequence.

After all bytes in the register have been transmitted to the accessory, the Auth 3.0 CP will return 0x00 in response to all further reads until a new register number is selected.

68.5.7 Registers

68.5.7.1 Device Version

Name	Address	Block	Type	Power-Up Value	Bytes	Access
Device Version	0x00	0	uint8	0x07	1	Read-only

Reading from this register will return 0x07.

68.5.7.2 Authentication Revision

Name	Address	Block	Type	Power-Up Value	Bytes	Access
Authentication Revision	0x01	0	uint8	0x01	1	Read-only

Reading from this register will return 0x01.

68.5.7.3 Authentication Protocol Major Version

Name	Address	Block	Type	Power-Up Value	Bytes	Access
Authentication Protocol Major Version	0x02	0	uint8	0x03	1	Read-only

Reading from this register will return 0x03.

68.5.7.4 Authentication Protocol Minor Version

Name	Address	Block	Type	Power-Up Value	Bytes	Access
Authentication Protocol Minor Version	0x03	0	uint8	0x00	1	Read-only

Reading from this register will return 0x00.

68.5.7.5 Device ID

Name	Address	Block	Type	Power-Up Value	Bytes	Access
Device ID	0x04	0	uint8	0x00 0x00 0x03 0x00	4	Read-only

Reading from this register will return 0x00 0x00 0x03 0x00.

68.5.7.6 Error Code

Name	Address	Block	Type	Power-Up Value	Bytes	Access
Error Code	0x05	0	uint8	0x00	1	Read-only

Table 68-16 Error Code register parameters

Bit	Name	Description
7-0	Error Code	See Table 68-17 (page 609).

Table 68-17 Error Code values

Value	Description
0x00	No error.
0x01	Reserved
0x02	Invalid register specified or register is read-only.
0x03-0x04	Reserved
0x05	Sequence error, command out of sequence.
0x06	Internal process error during challenge response generation.
0x07-0xFF	Reserved

Reading the Error Code register will clear it.

Attempts to access an undefined register may result in an error code of 0x02.

68.5.7.7 Authentication Control and Status

Name	Address	Block	Type	Power-Up Value	Bytes	Access
Authentication Control and Status	0x10	1	uint8	0x00	1	Read/Write

When read from, the Authentication Control and Status register provides the status of the most recently requested Auth 3.0 CP process, as shown in [Figure 68-13](#) (page 610) and [Table 68-18](#) (page 610). When written to, the Authentication Control and Status register controls the start of the Auth 3.0 CP processes, as shown in [Figure 68-14](#) (page 611) and [Table 68-21](#) (page 611).

Figure 68-13 Auth 3.0 CP Authentication Control and Status register, read-only bits

7	6	5	4	3	2	1	0
Error Set	Procedure Results			0	0	0	0

Table 68-18 Authentication Control and Status read-only register parameters

Bit	Name	Description
7	Error Set	See Table 68-19 (page 610).
6-4	Procedure Results	See Table 68-20 (page 610).
3-0	Reserved	

Table 68-19 Error Set values

Value	Description
0	The Error Code register contains a value of zero.
1	The Error Code register contains a non-zero value. Both this bit and the Error Code register contents are cleared after the Error Code register is next read or on the next power-on.

Table 68-20 Procedure Results values

Value	Description
0	Most recent challenge response generation process did not produce valid results or no challenge response generation process has been requested.
1	Accessory challenge response successfully generated.
2-7	Reserved

Figure 68-14 Auth 3.0 CP Authentication Control and Status register, write-only bits

7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	Proc Control

Table 68-21 Authentication Control and Status write-only register parameters

Bit	Name	Description
7-1		Reserved
0	Procedure Control	See Table 68-22 (page 611).

Note: Bits 7-3 must be zero. The data written to the Authentication Control and Status register must be 0x01 to initiate challenge response computation.

Table 68-22 Procedure Control values

Value	Description
0	No operation. This control does nothing and always reports success (Error Set = 0; Procedure Results = 0).
1	Start new challenge response generation process.

68.5.7.8 Challenge Response Data Length

Name	Address	Block	Type	Power-Up Value	Bytes	Access
Challenge Response Data Length	0x11	1	uint16	0	2	Read-only

The Challenge Response Data Length register holds the length in bytes of the results of the most recent challenge response generation process. If a completed challenge response is stored in the output buffer of the Auth 3.0 CP, this register will return a value of 64. In all other cases it will return a value of zero.

68.5.7.9 Challenge Response Data

Name	Address	Block	Type	Power-Up Value	Bytes	Access
Challenge Response Data	0x12	1	uint8	Undefined	64	Read-only

If the Auth 3.0 CP has computed a challenge response, the challenge response itself will be returned in this register. If there is no valid challenge response stored in the Auth 3.0 CP SRAM, the interface will return 0x00 for all bytes.

68.5.7.10 Challenge Data Length

Name	Address	Block	Type	Power-Up Value	Bytes	Access
Challenge Data Length	0x20	2	uint16	0	2	Read-only

The Challenge Data Length register holds the length in bytes of the current challenge stored within the Auth 3.0 CP. If a challenge is valid within the Auth 3.0 CP it will return a value of 32, otherwise, it will return a value of zero on a read.

68.5.7.11 Challenge Data

Name	Address	Block	Type	Power-Up Value	Bytes	Access
Challenge Data	0x21	2	uint8	Undefined	32	Read/Write

The Challenge Data register holds the current challenge data. If a valid challenge is stored within the Auth 3.0 CP, its value may be read via this register, otherwise reads of this register return all zeros.

This challenge data must be written to the Auth 3.0 CP by the accessory and must always be 32 bytes in length. If any number of bytes other than 32 is sent to the Auth 3.0 CP, the challenge register will be cleared and an error returned.

68.5.7.12 Accessory Certificate Data Length

Name	Address	Block	Type	Power-Up Value	Bytes	Access
Accessory Certificate Data Length	0x30	3	uint16	607-609	2	Read-only

The Accessory Certificate Data Length register holds the length of the X.509 certificate that is stored within the Auth 3.0 CP. The length of a certificate varies between 607 and 609, inclusive.

68.5.7.13 Accessory Certificate Data

Name	Address	Block	Type	Power-Up Value	Bytes	Access
Accessory Certificate Data 1	0x31	3	uint8	Certificate	128	Read-only
Accessory Certificate Data 2	0x32	3	uint8	Certificate	128	Read-only
Accessory Certificate Data 3	0x33	3	uint8	Certificate	128	Read-only
Accessory Certificate Data 4	0x34	3	uint8	Certificate	128	Read-only
Accessory Certificate Data 5	0x35	3	uint8	Certificate	128	Read-only

The Accessory Certificate Data registers hold the binary X.509 certificate that is stored in Auth 3.0 CP. The accessory certificate may be read from the Auth 3.0 CP in 128-byte pages starting at any of Accessory Certificate Data addresses, or it may be read in a continuous stream starting at Accessory Certificate Data 1. The Accessory Certificate Data Length value can be read to determine the exact number of bytes in the Accessory Certificate Data.

The accessory must wait at least t_{CERT} for the Auth 3.0 CP to create the certificate from the stored elements whenever it is requested. This means that if any register within block 3 is selected via a zero length write, the subsequent read operation must not be issued for t_{CERT} , even if the only intention is to read the length register.

68.5.7.14 Self-Test Status

Name	Address	Block	Type	Power-Up Value	Bytes	Access
Self-Test Status	0x40	4	uint8	0x00	1	Read-only

The Self-Test Status register reports the results of the X.509 certificate and private key tests, as shown in [Figure 68-15](#) (page 613) and [Table 68-23](#) (page 613).

Figure 68-15 Auth 3.0 CP Self-Test Status register

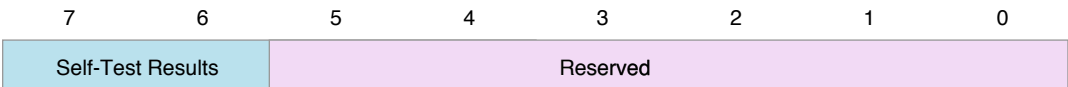


Table 68-23 Self-Test Status register parameters

Bit	Name	Description
7	X.509 Certificate	See Table 68-24 (page 613).
6	Private key	See Table 68-24 (page 613).
5-0	Reserved	

Table 68-24 Self-Test Status values

Value	Description
0	Error
1	Pass

Note: The X.509 and private key tests only verify that these elements are present in flash memory; no authentication is performed.

68.5.7.15 Device Certificate Serial Number

Name	Address	Block	Type	Power-Up Value	Bytes	Access
Device Certificate Serial Number	0x4E	4	uint8	Certificate	32	Read-only

The Device Certificate Serial Number register returns the certificate serial number as an ASCII string. Each byte in the serial number stored within the device will be returned as two ASCII characters in the range of 0-9, A-F.

68.5.7.16 Sleep

Name	Address	Block	Type	Power-Up Value	Bytes	Access
Sleep	0x60	1	uint8	Undefined	1	Write-only

Writing 0xEE to this register will force the Auth 3.0 CP to sleep.